

Table 1 | Electricity measurements and monitoring boundaries

Area	Provider and native resolution	Accounting boundary
Germany–Luxembourg	SMARD / hourly energy	Grid load excludes pumping and PV self-consumption
France	RTE eCO2mix / 30 minutes	Metropolitan area excluding Corsica; network losses included, pumping excluded
Belgium	Elia / 15 minutes	Control-area load with distributed estimation; Sotel overlap retained
DK1 and DK2	Energinet settlement / hourly energy	Separate bidding areas; self-consumption overlap corrected; all reported wind/PV retained